SUSTAINABLE REAL ESTATE

Regulatory Alignment · CRREM Pathways · Quick-Win Strategies

A Briefing for Real Estate Investors, Asset Managers & Property Owners

40%	**2024**	**+11%**	**$5T+**
of global CO_2 from buildings	year most assets exceed 1.5°C pathway	green rent premium offices	annual investment to decarbonise RE

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1 · EXECUTIVE SUMMARY

KEY FINDING: The majority of institutional real estate assets globally are already operating above the CRREM 1.5°C pathway threshold. Without structured decarbonisation programmes, up to 30% of prime office supply globally will fail to meet net-zero occupier demand by 2030. The cost of inaction now materially outweighs the cost of action.

Buildings account for approximately 40% of global final energy consumption and nearly 38% of energy-related CO_2 emissions. Against this backdrop, a structural convergence of regulatory pressure, investor ESG expectations and tenant demand for low-carbon space is reshaping the economics of real estate ownership across every major market. The window for orderly, cost-effective decarbonisation is narrowing, and the financial penalties for delay — in the form of brown discounts, regulatory fines and refinancing restrictions — are becoming measurable and material.

This report is structured around three interconnected themes. First, it maps the regulatory landscape across the EU, UK and US, identifying the compliance deadlines and standards that are most immediately consequential for real estate portfolios. Second, it explains how the Carbon Risk Real Estate Monitor (CRREM) — the industry's most widely adopted science-based benchmarking tool — translates the Paris Agreement into asset-level carbon intensity targets, and quantifies the performance gap between current stock and where it needs to be. Third, it presents five high-impact strategies that investors, asset managers and property owners can implement within 6–18 months to improve 1.5°C pathway alignment, reduce operational carbon and access green finance. Throughout, data and evidence are drawn from CRREM, CBRE, JLL, RICS, GRESB and leading proptech platforms including Deepki, Optiml, Scaler and BuildingMinds.

The overarching message is unambiguous: real estate aligned with CRREM pathways and the major regulatory frameworks push asset managers to act upon their sustainability strategies and may support attracting institutional capital yet **the green premium and brown discount is evidently driven by ESG-committed, institutional tenants and the resilience of an asset against climate risk.** Real estate that fails to act faces an accelerating convergence of value erosion, increasing operational expenses and tenant flight.

2 · REGULATORY LANDSCAPE

2.1 Compliance Urgency by Jurisdiction

The heatmap below provides a rapid-read assessment of where compliance pressure is most acute. Areas rated HIGH demand portfolio action in the current or next financial year — deferral carries direct financial consequences including fund reclassification, regulatory fines and restricted refinancing.

Regulatory Area	🇪🇺 EU	🇬🇧 UK	🇺🇸 US
Carbon / GHG Disclosure	HIGH	Medium	Medium
ESG Fund Classification (SFDR/SDR)	HIGH	HIGH	Low
Corporate Sustainability Reporting	HIGH	Medium	Medium
Minimum Energy Standards (MEES/BPS)	Medium	HIGH	HIGH
EU Taxonomy Alignment	HIGH	Low	—
EPBD — Building Renovation Plans	HIGH	Medium	—
Building Carbon Caps (LL97/BPS)	Low	Medium	HIGH
EPC / Energy Label Reform	Medium	HIGH	Low

Urgency: ■ Low ■ Medium ■ HIGH — action required this financial year

2.2 Key Milestones: 2024–2030

The table below consolidates the most consequential regulatory deadlines across the three major jurisdictions. Many of these are already in force; several carry direct asset-level consequences — from caps on lettable carbon intensity to mandatory sustainability disclosures in annual reports and fund prospectuses.

Year	Framework	Key Requirement	Scope
2024	**CSRD Wave 1**	NFRD-scope companies report FY2024 data under ESRS; real estate entities must disclose Paris-alignment strategy and transition plan	🇪🇺 EU
2024	**SFDR Level 2**	Full product-level PAI disclosures mandatory; Article 8/9 fund classification enforced; CRREM pathway alignment strongly recommended	🇪🇺 EU
2024	**NYC Local Law 97**	Carbon caps live for buildings >25,000 sq ft; ~8% non-compliant in first year; rises to ~50% by 2030 without intervention (Urban Green Council)	🇺🇸 US

2025	**CSRD Wave 2**	All large EU undertakings (+500 employees) report FY2025; EU Taxonomy alignment extends to all six environmental objectives from 2026	🇪🇺 EU
2025	**California SB 253**	$1B+ revenue companies doing business in California disclose Scope 1/2 emissions; Scope 3 required from 2027	🇺🇸 US
2025	**UK SDR (FCA)**	Four investment labels (Sustainability Focus / Improvers / Impact / Mixed Goals) operational for UK-authorised funds	🇬🇧 UK
2026	**EU Omnibus Package**	CSRD scope proposed to narrow to >1,000 employees (−80% entities); SFDR revised toward three product labels with 70% qualifying-asset threshold	🇪🇺 EU
2026	**UK EPC Reform**	Multi-metric EPC framework: Fabric Performance Metric + Heating System or Smart Readiness Metric replaces single Energy Efficiency Rating (EER)	🇬🇧 UK
2027	**EU EPBD — Renovation**	Worst-performing non-residential buildings (EPC F/G) must begin renovation; national Building Renovation Plans with binding milestones	🇪🇺 EU
2027	**UK MEES — EPC C**	All commercial lets must achieve EPC C (proposed); ~60% of Central London office stock currently below threshold (CBRE 2025)	🇬🇧 UK
2028	**CSRD — Non-EU entities**	Non-EU companies with €150M+ EU turnover must comply; global REITs and managers with European exposure directly in scope	🇪🇺 EU
2030	**UK MEES — Residential**	All PRS homes must achieve EPC C by 1 October; fines up to £30,000 per breach per property; Warm Homes Plan backed by £15B public funding	🇬🇧 UK
2030	**UK MEES — Commercial EPC B**	Commercial properties must achieve EPC B (deadline likely 2030–2035); 58% of Central London office sq ft currently falls short (CBRE)	🇬🇧 UK
2030	**US State BPS Network**	15+ jurisdictions have Building Performance Standards in force; 50+ cities have mandatory energy disclosure laws; carbon pricing expansion expected	🇺🇸 US

2.3 EU Regulatory Architecture

The European Union has constructed the world's most comprehensive sustainable finance regulatory architecture. Four interlocking instruments — EU Taxonomy, SFDR, CSRD and the recast EPBD — together define what constitutes a sustainable building, what must be disclosed about it, and who is required to act. Understanding how these frameworks interact is essential to building a coherent compliance strategy.

The EU Taxonomy provides the foundational classification system, identifying which economic activities qualify as environmentally sustainable across six objectives. For real estate, the most material are Climate Change Mitigation (CCM) and Climate Change Adaptation (CCA). To qualify under the CCM objective, an asset must either achieve an EPC of A or fall within the top 15% of the national building stock by primary energy demand. From 2026, alignment reporting expands to the four remaining objectives — Water, Circular Economy, Pollution, and Biodiversity — adding material new data collection requirements for portfolio owners and managers.

SFDR operationalises the Taxonomy at the fund level, requiring financial market participants — including real estate fund managers — to classify their products and disclose how sustainability characteristics are integrated. Article 8 funds must demonstrate ESG promotion; Article 9 funds must evidence sustainable investment as the primary objective. In November 2025, the European Commission proposed replacing the current Article 6/8/9 classification with three product labels (ESG Strategy, Sustainable Impact and Transition), each requiring a 70% minimum threshold for qualifying assets. Separately, entity-level Principal Adverse Impact (PAI) disclosures are proposed for removal, intended to reduce duplication with CSRD — a welcome simplification for most asset managers.

The CSRD mandates finance-grade sustainability disclosures using European Sustainability Reporting Standards (ESRS). For the real estate sector, ESRS E1 (Climate Change) is the critical standard: it requires companies to disclose how their portfolios align with a 1.5°C pathway — making CRREM alignment effectively a regulatory input rather than a voluntary best practice. The CSRD's reporting cascade (Wave 1 for FY2024; Wave 2 for FY2025; non-EU entities from FY2028) means that a significant portion of the global institutional real estate market is already inside the reporting perimeter or will be shortly. The 2025 Omnibus package, which proposes narrowing CSRD scope to companies with more than 1,000 employees, reduces the number of entities directly captured — but does not diminish the underlying data and decarbonisation obligations for those that remain in scope.

The recast EPBD (May 2024) completes the picture at the asset level. It requires all new buildings to meet Zero Emission Building standards from 2028 (public buildings from 2026), introduces Renovation Passports providing building-specific long-term roadmaps, and mandates national renovation plans that prioritise worst-performing buildings. Member States must publish these plans by 2026, with binding intermediate milestones through 2030 and 2035.

2.4 United Kingdom

The UK has diverged from the EU framework post-Brexit while maintaining broadly equivalent ambition. The Minimum Energy Efficiency Standards (MEES) represent the most immediate compliance lever for commercial landlords.

Sector	Now (2023–)	2027 (proposed)	Oct 2029	Oct 2030+
Commercial	Min. EPC E (all leases)	**Min. EPC C**	New HEM:EPC mandatory	**Min. EPC B target**
Residential (PRS)	Min. EPC E	EPC reform underway	EPC C under old metric = grandfathered	**Min. EPC C required**

CBRE analysis (2025) estimates that approximately 58% of Central London office stock by square footage currently sits below EPC B. With the 2027 EPC C intermediate milestone approaching and the EPC B deadline likely between 2030 and 2035, landlords face an imminent capital expenditure imperative. Those who have not commenced systematic retrofit planning risk competing for constrained contractor capacity in a tightening market.

The UK government's Warm Homes Plan, published on 21 January 2026, mandates EPC C for all private rented homes in England and Wales by 1 October 2030, backed by up to £15 billion in public funding. The new multi-metric EPC framework — integrating a Fabric Performance Metric alongside a Heating System or Smart Readiness Metric — replaces the legacy single-metric EER from 2029. Properties achieving EPC C under the current methodology before October 2029 are grandfathered, creating an incentive for early action. At the fund level, the FCA's Sustainability Disclosure Requirements (SDR), introducing four investment labels from 2024/2025, align broadly with SFDR in spirit, ensuring UK-domiciled funds face comparable transparency and labelling obligations.

2.5 United States

The US federal landscape is more fragmented than the EU or UK, and has become further bifurcated since 2025. The SEC's March 2024 climate disclosure rules — which would have mandated Scope 1/2 GHG disclosures for large public companies — have been subject to a voluntary stay since April 2024 pending Eighth Circuit review, with their future under the current administration remaining uncertain. However, the absence of federal mandates has accelerated state-level action significantly.

New York City's Local Law 97, which came into force in 2024, represents the most aggressive asset-level carbon standard globally. Carbon caps covering all buildings above 25,000 square feet escalate in stringency through 2030 and beyond, with the Urban Green Council estimating that approximately 50% of covered buildings will be non-compliant by 2030 without material retrofit investment. California's SB 253 requires Scope 1 and 2 disclosure from 2026 and Scope 3 from 2027 for companies with $1 billion or more in revenue doing business in the state — a broadly applicable threshold that captures many real estate operating companies. Massachusetts, Colorado, Illinois and eleven further jurisdictions have enacted Building Performance Standards, and fifty cities have mandatory energy disclosure laws in force. For global portfolios with US exposure, the ISSB S2 standard, adopted or in adoption by 36 jurisdictions as of mid-2025, is increasingly becoming the de facto international baseline for climate disclosure.

3 · CRREM PATHWAY BENCHMARKS

3.1 Framework Overview

The Carbon Risk Real Estate Monitor (CRREM) is the real estate industry's most widely adopted science-based decarbonisation framework. Originally developed under the European Commission's Horizon 2020 programme and now funded by the Laudes Foundation alongside major institutional investors including APG, PGGM, NBIM and GPIF, CRREM translates the Paris Agreement's 1.5°C and 2°C carbon budgets into asset-level decarbonisation trajectories covering 28 countries, multiple property types and both commercial and residential sectors. Its pathways are embedded in GRESB scoring, recommended by the IIGCC Net Zero Investment Framework, and increasingly mandated through CSRD ESRS E1 disclosures. A 2023 GRESB survey found that over 70% of respondents with EU assets were either using CRREM or planning to implement it within twelve months.

CRREM measures performance against two complementary intensity metrics: carbon intensity (kgCO_2e per m² of gross lettable area per year) and energy intensity (kWh/m²/yr). An asset becomes misaligned with the 1.5°C pathway when its measured or modelled carbon intensity exceeds its applicable threshold — entering what the industry terms the carbon budget overshoot zone. According to GRESB data covering 150,000 assets, the majority of global institutional real estate is already operating above the 1.5°C pathway. Once assets exceed the pathway, they face a compounding set of consequences — regulatory penalties, restricted refinancing, loss of ESG-committed tenants and capital value erosion — that grow more severe with each year of inaction. The updated V2 pathway (2023, revised 2024) reflects the sector's aggregate underperformance versus original 2018 projections, meaning pathways have become steeper across most geographies and property types.

3.2 Carbon Intensity Targets by Asset Type (1.5°C Scenario)

The table below presents indicative CRREM carbon intensity benchmarks across major asset types. Values vary materially by geography, reflecting differences in grid carbon intensity, climate conditions, building stock characteristics and national policy ambitions. France, with its predominantly nuclear electricity supply, faces substantially lower near-term targets than Germany or the United States, where fossil fuels remain more significant in the energy mix.

Asset Type	2024 Actual	2025	2030	2040	2050	Total Cut
Office (EU avg.)	28	26	18	9	≤3	**~89%**
Retail / Shopping	32	30	20	10	≤3	**~91%**
Logistics / Ind.	18	17	11	5	≤2	**~89%**
Residential (EU)	22	20	13	6	≤2	**~91%**
Office (UK)	30	28	19	10	≤3	**~90%**

Office (US avg.)	40	38	26	13	≤4	**~90%**
Hotel / Hospitality	55	52	34	17	≤5	**~91%**

All values in $kgCO_2e/m^2/year$ (1.5°C scenario, mid-range). Source: CRREM Foundation 2024/2025. Grid decarbonisation assumptions embedded per country.

3.3 Current Performance Gap — Assets Already Exceeding the Pathway

The chart below illustrates the estimated share of assets reported to the GRESB 2024 benchmark cycle that currently exceed their applicable CRREM 1.5°C carbon intensity threshold. Hotels and suburban offices face the most acute exposure; data centres, despite their high absolute energy consumption, tend to perform better on a per-m² basis partly due to their electrified energy profiles and efficiency investments driven by operational cost pressures.

Sector	Rationale	Share of Assets Exceeding CRREM 1.5°C
Hotel / Hospitality	Very high electric, heating and water demand	**79%**
Suburban Office	Lower value vs. CBD decreases ROI on efficiency CapEx	**74%**
Retail / Shopping Centre	High energy demand, operating hours and large spaces	**71%**
CBD/ Prime Office	High heating & cooling and ventilation demand	**68%**
Residential (PRS / Multifamily)	Varied energy efficiency depending on specs	**62%**
Logistics / Industrial	Low heating demand, electrified with on-site energy potential	**55%**
Data Centre	Most recent asset class, fully electrified and efficient	**48%**

Source: GRESB 2024 Benchmark Report; CRREM Foundation analysis. Figures are estimates based on reported energy/carbon intensity data vs. CRREM 1.5°C pathway thresholds.

3.4 The Green Premium and the Brown Discount

The financial consequences of CRREM alignment — or misalignment — are increasingly visible in transaction pricing, rental markets and financing conditions. The green premium describes the uplift in rents and capital values commanded by high-performing, energy-efficient assets. The brown discount describes the corresponding penalty applied to inefficient properties, which is intensifying as regulatory deadlines approach and occupier ESG requirements tighten.

JLL's research consistently identifies rental premiums of 7–11% for green-certified offices relative to non-certified peers, driven primarily by occupier ESG commitments and the operating cost savings that energy-efficient space delivers. CBRE's European Investor Intentions Survey (2024) found that 32% of institutional investors are willing to pay a premium for ESG-aligned assets, with more than half of those prepared to pay in excess of 20%. The CBRE UK Sustainability Index demonstrates statistically significant total return outperformance of up to 18.2% in the

Netherlands and the UK for assets with EPC A or B ratings compared to lower-rated stock. JLL's Return on Sustainability report identifies 5–15% valuation discounts for inefficient assets in markets with active minimum energy performance standards in force — discounts that are expected to deepen materially as the MEES regulatory escalator moves from EPC E to C to B between now and 2030.

EPC Rating (UK) Commercial Office Market	Efficiency measures	Rental Premium / Discount
EPC A (Best-in-Class)	No CapEx required for years, attracting institutional grade tenants	**+11%**
EPC B (High Performance)	Low-hanging measures and end-of-life replacements	**+7%**
EPC C (Market Baseline)	Smart upgrades; EPC C expected regulatory minimum in 2027	**Baseline**
EPC D (Below Standard)	Significant upgrades required	**−8%**
EPC E (Legal Minimum)	Major CapEX measures	**−15%**
EPC F/G (Non-Compliant)	Full retrofit to avoid distressed (no new leases/renewals allowed)	**−22%+**

Sources: JLL Return on Sustainability (2023); CBRE UK Sustainability Index Q4 2024; Building Atlas/VOA analysis (2025). Premiums and discounts are relative to EPC C baseline.

4 · COST OF ACTION VS. INACTION

The financial case for proactive decarbonisation is now stronger than the case for delay across every material dimension. The table below structures the comparison across five key risk areas, drawing on published research from JLL, CBRE, GRESB and the Urban Green Council.

Risk Dimension	✓ Act Now	✗ Do Nothing
Asset Value	Green premium: +6–12% rents; +20% investor willingness (CBRE 2024)	Brown discount: −5–22% valuation; stranded asset write-downs (JLL 2023)
Regulatory	CSRD/SFDR compliant; avoids fines; qualifies for green finance	NYC LL97 fines; UK MEES penalties up to £150,000/property
Financing	15–30 bps SLL margin benefit; green bond eligibility	Lenders restricting refinancing for sub-benchmark assets
Tenant Retention	Meets SBTi/net-zero occupier needs; lower vacancy; faster lease-up	Losing ESG-committed tenants; rising void periods; income erosion
Long-term Value	CRREM-aligned = investable 2030+; attracts institutional mandates	Majority of GRESB assets already above 1.5°C pathway (2024); accelerating obsolescence

5 · TOP 5 QUICK-WIN RECOMMENDATIONS

The five actions below are selected for their high impact-to-effort ratio and their compatibility with the regulatory and CRREM frameworks described above. Each can be initiated within the current financial year and is expected to produce measurable results within 6–18 months. Together, they form the minimum viable decarbonisation programme for any portfolio managing material transition risk.

#	Action	Expected Impact	Effort / Cost	Tools / Partners
1	**CRREM Gap Analysis & Pathway Alignment Register**	1.5°C pathway deviation per asset; capex triage; satisfies CSRD ESRS E1, SFDR PAI 18, GRESB data needs	Low / No capex	Deepki · Optiml · CRREM XLS
2	**IoT Sub-Metering & Real-Time Monitoring**	10–20% energy reduction within 12 months; measured (not estimated) emissions baseline	Med / €15–40/m²	Scaler · BuildingMinds · Siemens
3	**LED + HVAC Optimisation & Fabric Upgrades**	15–30% energy saving; immediate EPC uplift; payback 2–5 yrs (LED/HVAC); fabric supports UK HEM:EPC	Med / Asset-specific	RICS-accredited auditors
4	**Green Lease Clauses & Tenant Data-Sharing**	Removes split-incentive barrier; unlocks whole-building data; 6–12% occupant energy reduction (JLL)	Low / Legal costs only	BBP Model Green Lease · BOMA
5	**Green Finance / Sustainability-Linked Loans**	15–30 bps margin benefit; funds retrofit; SFDR Art.9 eligibility; signals ESG credibility to market	Low / Structuring fee	ING · LBBW · NatWest · LMA

5.1 CRREM Gap Analysis & Pathway Alignment Register

The most fundamental — and most frequently deferred — action is the systematic mapping of every asset in a portfolio against its applicable CRREM pathway. This exercise calculates each asset's current 1.5°C pathway deviation and projects the year at which it exceeds the threshold under a business-as-usual scenario, producing an actionable risk register that drives capital allocation decisions. Practically, it can be completed in four to twelve weeks using platforms such as Deepki, Optiml or the native CRREM XLS Tool. The outputs — pathway gap, required annual intensity reduction, intervention cost estimate — directly satisfy the data requirements of CSRD ESRS E1, SFDR PAI 18 and the GRESB Performance Component. The analysis should also inform acquisition due diligence: assets already above the pathway or with near-term threshold crossings before 2030 should carry an explicit transition risk premium in any financial model.

5.2 IoT Sub-Metering & Real-Time Monitoring

A significant proportion of ESG data used in real estate reporting is estimated from benchmark assumptions rather than measured from actual consumption. This is both an accuracy problem — estimated baselines undermine retrofit targeting and CRREM benchmarking — and a credibility problem, creating exposure to greenwashing risk under the incoming CSRD and SFDR disclosure regimes. Deploying IoT sub-metering at circuit or zone level across electricity, gas, heating and cooling systems establishes the measured performance baseline required for credible pathway alignment, and generates the anomaly detection capability needed to capture operational savings. Evidence from UK and Irish PBSA portfolios using similar platforms shows 10–20% energy reduction achievable within twelve months of deployment through automated load scheduling, occupancy-responsive controls and real-time fault detection. The incremental cost — typically €15–40 per m² of installed capacity — is routinely recoverable within two to four years through energy bill savings alone.

5.3 LED Lighting, HVAC Optimisation & Fabric Improvements

For most existing commercial buildings, the fastest path to CRREM pathway compliance and improved EPC ratings runs through three proven operational and fabric interventions. LED lighting retrofits typically deliver 50–70% lighting energy reductions with payback periods of two to four years and minimal operational disruption. HVAC recommissioning — including variable speed drives on pumps and fans, BMS optimisation and smart thermostatic controls — can reduce heating and cooling energy consumption by 15–25% with payback periods of two to five years. Fabric improvements, including enhanced roof and wall insulation, high-performance glazing and thermal bridging remediation, address the structural energy demand of a building and are essential for satisfying the UK's new Fabric Performance Metric under the HEM:EPC framework from 2029. RICS advises adopting a fabric-first sequencing approach — reducing total energy demand before switching fuel sources — to avoid the counterintuitive outcome where heat pump installation reduces the EPC rating under legacy cost-based methodology, despite delivering genuine carbon reductions.

5.4 Green Lease Clauses & Tenant Data-Sharing

The structural barrier most consistently cited as delaying commercial real estate decarbonisation is the split incentive: landlords bear the cost of building improvements but tenants capture the savings through lower energy bills. Green lease clauses directly address this misalignment by embedding mutual sustainability obligations into the legal contract governing the landlord-tenant relationship. At minimum, green leases should include provisions for quarterly whole-building metered data disclosure, HVAC and lighting operational standards, tenant fit-out sustainability specifications (prohibiting gas-fired appliances, mandating LED and low-energy equipment), and a joint sustainability improvement plan with annual review and accountability. JLL's research shows that tenants covered by green leases demonstrate 6–12% lower energy consumption than those under conventional leases, driven primarily by heightened awareness and shared monitoring accountability. Template frameworks are available from the Better Buildings Partnership (UK), BOMA (US) and GRESB, all of which have been updated to reflect current regulatory requirements. Legal drafting costs are typically the only material upfront expense.

5.5 Green Finance & Sustainability-Linked Loan Refinancing

The green finance market has grown dramatically in scale and sophistication: global green bond issuance exceeded €500 billion in 2024, with real estate representing more than 20% of use-of-proceeds volumes. Sustainability-linked loans (SLLs) tied to CRREM pathway performance milestones, GRESB score improvement thresholds or specific EPC rating targets offer margin step-downs of 15–30 basis points for meeting agreed sustainability KPIs, with corresponding step-ups for non-compliance. This cost-of-capital benefit is not the primary driver — at current loan sizes, a 20 basis point margin reduction on a €100 million facility saves €200,000 annually, which is material but secondary to the asset value and regulatory risk benefits described above. The more significant strategic advantages of green finance are the credibility signal it sends to institutional co-investors and joint venture partners, its direct contribution to SFDR Article 9 fund eligibility, and its role in pre-empting the refinancing restrictions that lenders are increasingly imposing on sub-benchmark carbon-performing assets.

6 · PROPTECH & DATA PLATFORM LANDSCAPE

The delivery of any credible decarbonisation programme depends entirely on the quality of underlying asset data. Every action described above — from CRREM gap analysis to green lease design to SLL structuring — requires accurate, granular, auditable energy and carbon data at the asset and portfolio level. A growing ecosystem of proptech platforms has emerged to supply this infrastructure, each with a distinct approach, ownership structure and client positioning.

All platforms reviewed here align with CRREM pathways and the major regulatory disclosure frameworks; that is table stakes in the institutional market. The meaningful differentiators lie in technical architecture, financial modelling depth, data collection methodology, geographic coverage and the maturity of each platform's underlying dataset. The table below provides a structured comparison across these dimensions, including ownership structure — an important consideration for vendors entering long-term data management relationships — and representative clients.

Platform	Key Differentiators	Limitations	Client Examples
BuildingMinds **Corporate spin-off** (Schindler Group) Berlin · est. 2018	• API-first Common Data Model — any system integrates seamlessly, eliminating portfolio data silos • Digital Building Twin enables predictive maintenance and long-term scenario modelling • Ability to extract tenancy agreement data to review financial and timelines to align with CapEx planning	Weaker IRR/NPV modelling; retrofit planning	Zurich Insurance General · EQT RE · AXA IM Alts · Brookfield · GOLDBECK · La Française RE
Deepki **VC-backed** (Series C, €150M) One Peak · Highland Europe · Bpifrance	• World's largest RE ESG database (€4T+ AUM, 80+ countries) — AI benchmarking accuracy unmatched by smaller platforms • ISAE 3000 certified: audit-ready data at financial reporting grade — essential for CSRD, SFDR and lender due diligence • AI Virtual Retrofit identifies energy measures via thermodynamic modelling without physical surveys	Complex onboarding; pricing for large institutions; lighter on NPV/IRR modelling	PGIM · Allianz RE · AEW · Tikehau · Warburg HIH · French Govt.
EVORA / SIERA **PE/Institutional** (Bridges · MSCI · Farview) UK · est. 2011 · Independent	• Only platform integrated within a 200+ person consultancy — software outputs include expert human interpretation • ~20% of European GRESB submissions and ~10% globally supported in 2024 — unmatched methodology depth • Acquired Metry (2024) for automated Nordic utility data ingestion	Software bundled with advisory; less suited to pure SaaS buyers; primarily UK/Europe focus; smaller tech team.	Tishman Speyer · Hines · Invesco RE · M&G · BNP Paribas RE · Schroder RE
IBM Envizi **Corporate subsidiary** (IBM, NYSE: IBM) Acquired 2022 Enterprise division	• Verdantix Green Quadrant Leader 2025 — top-rated across 21 providers for data acquisition and AI-assisted GHG calculations • Global JLL partnership (2024) makes Envizi the ESG data backbone for thousands of JLL-managed properties • Most rigorous data governance reviewed: audit trails, confidence scoring and human review steps throughout	Not purpose-built for real estate; CRREM modelling needs third-party integration; enterprise pricing unsuited to mid-market.	JLL (global partner) · IBM Global RE · Ikano Group · Downer Group · Growthpoint Properties AU

Measurabl **VC-backed** (Series D, $93M) Energy Impact Partners Sway · S&P Global Salesforce Ventures	• Quantum Cloud: 18B sq ft, $3T AUM, 93 countries — peer benchmarking at a scale no other platform matches • ISO 14064-3 verified + ENERGY STAR Partner of the Year 6 years running • US BPS compliance module auto-calculates regulatory fines and generates state-specific retrofit plans	US-centric heritage; EU regulatory coverage less mature than European platforms; less sophisticated retrofit IRR/NPV	Boston Properties · Lincoln Property Co. · Colliers · The Social Hub ·
Optiml **VC-backed** (Pre-Seed+, $15M) BitStone Capital KOMPAS VC Innovation Endeavors Planet A Ventures	• ETH Zurich algorithms link CRREM pathway analysis to Capex, IRR and NPV — the only platform producing investment-committee-ready business cases • Works with sparse data (building type, area, location sufficient) — accessible for assets with incomplete records during due diligence • 90% time reduction vs. manual analysis — enables scalable strategy delivery across large portfolios	Earlier-stage product; lighter automated data collection vs. Deepki; Scope 3 embodied carbon still developing.	PATRIZIA · Empira · Pension Fund BEWAG · Longevity Partners · pom+ Consulting
Scaler **VC-backed** (Series A, $10M) Plural Base10 Partners est. 2022 · NYC/Amsterdam	• Lowest adoption barrier: single interface serves CFOs, sustainability leads and property managers without specialist training • ML anomaly detection flags underperforming assets in real time — shifts from periodic reporting to continuous monitoring • Clients reduced CO_2 output by 15M kg annually — strong documented outcomes relative to platform maturity	Smallest team (~50 staff); limited retrofit financial modelling; fewer enterprise integrations; narrower multi-jurisdictional coverage.	AlteraA · Dream · Vesteda · CLS · The July · DW Property · M Multi

Note: All platforms reviewed support CRREM pathway benchmarking, GRESB data integration, and the principal regulatory disclosure frameworks (CSRD, SFDR, EU Taxonomy). The table focuses on differentiating attributes. Ownership and client data as of April 2026.

Several notable dynamics are reshaping the competitive landscape. Deepki remains the market-share leader by AUM monitored (€4 trillion, 80+ countries) and has moved decisively toward offering an end-to-end ESG platform following its acquisition of embodied carbon SaaS Nooco and its partnership with PGIM Real Estate and CBRE. Optiml, by contrast, differentiates on financial depth: its ETH Zurich-derived algorithms generate scenario-specific capex, IRR and NPV outputs that allow asset managers to build credible investment committee cases for retrofit expenditure — a capability that pure-play reporting platforms do not replicate. The BuildingMinds–Optiml strategic partnership announced in March 2025 addresses this gap directly, combining BuildingMinds's data aggregation and digital twin strengths with Optiml's retrofit scenario modelling. Scaler, the youngest and most focused of the seven, has positioned itself as the operationally accessible option for teams where sustainability is managed across multiple functions rather than by a dedicated ESG team.

Of the three additions to this landscape, IBM Envizi stands apart by scale of enterprise pedigree: its July 2024 partnership with JLL — under which Envizi now underpins JLL's global Sustainability Program Management offering — effectively makes it the back-end data engine for thousands of JLL-managed commercial properties worldwide.

However, its cross-sector heritage means it lacks the real estate-specific CRREM pathway modelling and retrofit financial analytics of purpose-built platforms. Measurabl is the dominant platform in North America: 18 billion square feet, $3 trillion in AUM across 93 countries, and 37% of the world's top asset managers — figures that no other platform approaches in scale. Its US BPS compliance tools and Decarb module for auto-generated retrofit plans fill a gap that European platforms have been slower to address. EVORA's SIERA occupies a unique position as the only platform reviewed here that is structurally integrated with a 200-strong consultancy: SIERA is both software and the data backbone for EVORA's advisory services, meaning clients receive expert interpretation alongside the platform output — a meaningful advantage in complex regulatory or 1.5°C pathway analysis situations where automated outputs alone are insufficient.

7 · CONCLUSION — THREE STRATEGIC IMPERATIVES

The transformation of sustainable real estate from an optional add-on to a baseline operational requirement is now structural and irreversible. The combination of tightening regulation, rising financing costs for non-compliant assets, and the accelerating divergence between green and brown rental markets has resolved the question of whether decarbonisation is financially justified. It has. The question now is one of sequencing and speed.

1 **Establish your data foundation first.** Without asset-level, metered and auditable energy and carbon data, CRREM benchmarking, regulatory compliance and green finance structuring are all impossible. Investment in IoT sub-metering and an ESG data platform is not a sustainability cost — it is the prerequisite for every subsequent decision.

2 **Sequence capital by 1.5°C/2.0°C pathway deviation, not convenience.** CRREM gap analysis produces a ranked risk register — ordered by degree of pathway misalignment — that should directly drive the capital expenditure plan. Assets furthest above the threshold, with the highest capital values and greatest retrofit feasibility, must be prioritised. Sustainability-linked loans and green bonds should fund the programme, simultaneously reducing cost of capital and establishing market credibility.

3 **Engage tenants through the lease, not just conversation.** Green lease clauses are the fastest available route to shared accountability, whole-building measured data and aligned incentives. Without them, the split-incentive problem will persist regardless of how committed the landlord's decarbonisation intentions are. Green lease adoption should be a condition of any new letting or lease renewal from this point forward.

Real estate that demonstrates measurable CRREM pathway alignment, holds relevant certification and is financed through green instruments will command the green premium and retain ESG-committed tenants. Critically, value is not driven by compliance alone — it is driven by the resilience of an asset against climate risk and its ability to meet the expectations of institutional tenants who increasingly define lettability. Real estate that defers action faces an accelerating convergence of value erosion, increasing operational expenses and tenant flight.

DISCLAIMER: This report is for informational purposes only and does not constitute investment, legal or financial advice. All regulatory timelines are subject to legislative change. Data and market figures are sourced from publicly available research and are indicative. Investors should seek qualified professional advice before acting on any information contained herein.

8 · KEY SOURCES & REFERENCES

Regulatory Frameworks

- EU Commission — CSRD, SFDR, EU Taxonomy, EPBD (2020–2024 incl. Omnibus Package Feb 2025)
- UK Government / DESNZ — Warm Homes Plan (Jan 2026); EPC Reform Partial Response (Jan 2026)
- FCA — Sustainability Disclosure Requirements and Investment Labels (2024)
- NYC Mayor's Office — Local Law 97 Carbon Caps (2024 Compliance Cycle)
- California Air Resources Board — SB 253 / SB 261 (2024/2025)

Research & Industry

- CRREM Foundation — Risk Assessment Reference Guide V2 (2024/2025); CRREM–SBTi Pathway Application (2025)
- CBRE — UK Sustainability Index Q4 2024; European Investor Intentions Survey 2024; MEES Analysis (June 2025)
- JLL — Return on Sustainability (2023); The Green Tipping Point (2024)
- GRESB — Real Estate Benchmark Report (2024); CRREM Integration and Stranding Year Analysis
- RICS — Sustainability and Commercial Property Valuation Professional Standards (2023)
- Urban Green Council — Three Real Estate Sustainability Risks to Manage in 2025 (Nov 2025)
- Optiml — Brown Discounts and Green Premiums: Carbon Performance and Valuations (2025)
- Building Atlas / Urban Neuron — The Hidden Cost of Inaction: Brown Discounts in CRE (2025)
- Jones Day — EPC C by 2030: What Real Estate Investors Need to Know (March 2026)
- European Real Estate Forum — EU Taxonomy Professional Standards Paper (Dec 2024)

Proptech Platforms (Data & Documentation)

- BuildingMinds — Schindler Group Parent Company Reporting; BuildingMinds–Optiml Partnership (March 2025); GRI Club PropTech of the Year 2024
- Deepki — Annual Growth Report 2025 (Feb 2025); Series C Announcement (March 2022); ISAE 3000
- EVORA Global / SIERA — World Green Building Council Profile; GRESB Partner Documentation; Bridges Fund Management Investment Announcement (Oct 2022); Metry Acquisition (2024)
- IBM Envizi — Verdantix Green Quadrant Leader 2025; JLL–IBM Partnership Announcement (July 2024); GRESB Partner Profile; IBM Newsroom
- Measurabl — Series D Announcement (May 2023); Next-Gen Platform Launch (July 2024); MIPIM 2026 Global ESG Compliancy Award; 2024 Year in Review (Feb 2025)
- Optiml — Client Win Announcements (Dec 2025); Pre-Seed Extension (Aug 2024); ETH Zurich Spin-off Documentation
- Scaler — Series A Announcement (July 2024); Platform Documentation (2024/2025)